

ADITI KALLE

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SUMMARY

Mechanical engineer specializing in thermal-fluids hardware for extreme environments: high-heat-flux cooling, two-phase and capillary thermal management, and the custom test rigs that qualify them. Designed and operated rigs reaching a critical heat flux of 600 W/cm^2 on capillary wicking structures, validating design parameters for cold plate and heat pipe architectures. Hands-on with ultra-high vacuum, high-pressure gas, and LOTO-governed test operations; refractory metal hardware; COMSOL transport modeling; and production Python analysis pipelines.

TECHNICAL SKILLS

Thermal Management: High-heat-flux design, cold plates, heat pipes, capillary wicking, multiphase and two-phase transport, boiling and dryout limits, conduction / convection / radiation sizing, COMSOL Multiphysics.

Test & Hardware Operations: Custom test rig design and fabrication, test stand operation, DAQ and high-density thermocouple instrumentation, ultra-high vacuum (UHV), high-pressure gas, LOTO and hazardous operations, test procedures and hardware acceptance criteria, root-cause analysis.

Materials & Diagnostics: Refractory alloys (tungsten, tantalum), porous metals, microstructure characterization (SEM/EDX), thermal analysis (TGA-DSC), surface coatings and wettability treatments, anti-fouling and scale dynamics.

Software & Analysis: Python (NumPy, SciPy, pandas, OpenCV, CVXPY), optimization (integer LP), time-series telemetry modeling, image processing, YAML-configured pipelines, CAD.

EXPERIENCE

Marine Energy Fellow (DOE / ORISE) — MarineSitu, Seattle, WA 2025 – Present

- **Power and energy budgeting for an unattended remote system:** characterized a 100 W solar / 1200 Wh LiFePO₄ platform from 35 days of telemetry, derived a ~230 Wh/day budget, and isolated true component loads (1.7 W base, 33.7 W camera) to set a 23% payload duty cycle as the governing constraint.
- **Sizing and forecasting:** built component-level power budgets across 7 operational profiles — battery 7 – 18 Ah, array 30 – 109 W, 141 – 783 Wh/day — supported by solar forecasting models (LSTM plus baselines, 0.93 W MAE).
- **Constrained scheduling and tooling:** formulated payload scheduling as an integer linear program (CVXPY) held inside a 20 – 90% state-of-charge envelope, with zero over-discharge events across the deployment; delivered two production Python tools with object-oriented, YAML-configured pipeline architecture.

Research Intern, Advanced Materials Group — Zap Energy, Seattle, WA 2024

- **Hardware qualification:** qualified multi-component refractory alloy systems and porous tungsten thermal management structures intended for extreme thermal environments and high-heat-flux service; performed TGA-DSC thermal analysis and SEM/EDX microstructural qualification.
- **Test definition and hazardous operations:** authored formal test setups, test procedures, and hardware acceptance criteria for high-stakes thermal hardware; executed testing under safety-critical UHV, high-pressure gas, and LOTO protocols.
- **Multiphysics simulation:** modeled porous-metal thermal-fluid transport in COMSOL — effective thermal conductivity, permeability, and pressure drop across severe temperature gradients; led cross-functional design reviews for working-fluid selection.

Graduate Research Assistant, Thermal Energy Materials & Physics Lab — UC San Diego 2022 – 2023

- **High-heat-flux test rig design:** custom-engineered and operated high-heat-flux thermal test rigs achieving a critical heat flux of 600 W/cm^2 on capillary wicking structures, validating key parameters for cold plate and heat pipe designs; configured DAQ and instrumentation and authored the rig safety assessments.
- **Analytical modeling:** modeled capillary pressure limits and thermal resistance using Young-Laplace and Darcy formulations, framing the dryout boundary as a pressure budget rather than a temperature limit.
- **Root-cause analysis:** led failure analysis on thermal-fluid anomalies (hydrostatic water-level drift, capillary stiction), separating instrumentation artifacts from physical effects and revising operating procedures to protect data integrity.

Visiting Scholar, Energy & Multiphase Flow Lab — University of Illinois Urbana-Champaign 2021 – 2022

- **Multiphase heat-transfer experiments:** ran saltwater falling-film evaporation on stainless-steel and 3D-printed polymer tubes and pool boiling on micron-scale grooved stainless samples, using an automated test system with DAQ, heater control, and high-speed imaging to quantify the trade between surface treatment and heat-transfer efficiency.
- **Characterization and data pipelines:** characterized CaCO₃ scale morphology by SEM; built a Python (OpenCV, pandas) image-processing and wave-velocity pipeline to quantify fouling-layer growth and liquid retention, supporting CFD model validation and manuscript preparation.

EDUCATION & PUBLICATIONS

M.S., Mechanical Engineering — University of California, San Diego GPA 3.61

B.Tech, Mechanical Engineering — PES University, Bengaluru, India GPA 3.6

Publications: Jin et al., “Falling-film evaporation in heat exchangers,” ASME J. Heat and Mass Transfer (2024) · Jin et al., “Anti-fouling surface treatments for solar-driven desalination,” Desalination (2023).